

PATH B — GREEN RETROFIT AUDIT

Feature Launch Kit

Green Audit Deck

Features ship. Departments find out late.

- Siloed teams, delayed comms, no single source of truth
- Sales, CS, Marketing and every dependent team scrambling to catch up
- Weeks of road shows and enablement sessions just to align

Launch Kit automates enablement materials so every team is ready on day one.

From POC to org-wide enablement engine.

POC TODAY

- Static knowledge base, 3 fixed doc types, single feature scope

AT SCALE

- RAG over full product catalog, multiple file types as input
- Outputs expand: PPTs, PDFs, battle cards, dev docs, web content and more
- Triggered automatically when a feature ships

One unit of value (R): One completed launch kit per feature release.

Small POC, manageable cost. Scaled org, compounding problem.

- 3 LLM calls per kit, one per doc type, every feature release
- Full knowledge base injected into every prompt every time
- No caching: same kit regenerated each time a user requests it
- At 20+ features per year across a product org, every inefficiency multiplies

Why believe this? No telemetry exists yet. Cost and token count are the proxies. Measurement is step one.

Honest limits before the solutions.

ASSUMPTIONS

Model	gpt-4o-mini used in POC (\$0.15/M input, \$0.60/M output). At scale, org selects model. Lighter alternatives (gpt-5 mini class) ran ~\$0.125/M input but output costs vary.
Call pattern	on-demand, no batching, no caching
Region	default OpenAI inference region, grid intensity unknown
KB size	small today, grows significantly at full product scale
Retention	outputs not stored, so repeat requests re-run the model

What we do not know yet: tokens per kit, calls per release cycle, actual energy per R.

All four pillars are in play.

HOTSPOT	CARBON	ENERGY	HARDWARE	MEASUREMENT
Repeated LLM calls, no cache	×	×		
Full KB in every prompt		×		
Single model for all doc types	×	×		
No usage telemetry				×
Idle infra between releases			×	

SOLUTION 1

Cache Kits Per Feature Release

PROBLEM

Every user request re-runs the model even for the same feature.

CHANGE

Store generated kits keyed by feature release ID. Serve from cache on repeat requests.

EXPECTED IMPACT

Eliminates duplicate LLM calls per feature. Reduces API cost per release cycle with no quality tradeoff.

PILLARS

Carbon

Energy

Pattern: Static Content (Green Software Patterns)

SOLUTION 2

RAG to Shrink Prompt Size

PROBLEM

As the KB grows to cover the full product catalog, injecting it whole into every prompt becomes expensive and wasteful.

CHANGE

Replace static injection with retrieval. Pull only the docs relevant to the current feature into the prompt.

EXPECTED IMPACT

Cost per kit stays flat as the product catalog grows. Token count per call does not scale with KB size.

PILLARS

Energy

Carbon

SOLUTION 3

Right-Size the Model Per Doc Type

PROBLEM

gpt-4o-mini runs for all 3 doc types regardless of complexity.

CHANGE

Route simpler outputs (customer FAQ, short email) to a lighter model. Reserve the larger model for complex outputs (sales battle card, solution brief).

EXPECTED IMPACT

Lower inference cost per doc type. Faster generation on simpler outputs reduces latency and API spend.

PILLARS

Carbon

Energy

| Caveat: Pair with a quality check on sampled outputs before auto-distribution. Not a blind downgrade.

SOLUTION 4

Batch Generation Over On-Demand Calls

PROBLEM

Docs are generated individually on user request, creating fragmented and repeated inference sessions.

CHANGE

Trigger full kit generation as a single batch job when a feature is marked ready. All doc types produced in one run.

EXPECTED IMPACT

Reduces idle inference time between calls. Kit is ready before any team has to request it, cutting time to enablement.

PILLARS

Hardware

Energy

Measurement Plan

WHAT TO TRACK FOR THE FIRST 2 WEEKS

- Tokens per kit (per R)
- LLM calls per feature release
- Cost per completed kit (cost as energy proxy)
- Cache hit rate once caching is implemented
- Latency per doc type (quality signal for model routing)

What counts as success: Cost per kit holds flat or decreases as feature volume scales.

Caveats & Honest Limits

- No live telemetry exists yet. Hotspot claims are reasoned estimates, not measured data.
- Carbon neutral is not claimed. SCI requires real grid intensity data we do not have.
- Offsets and RECs do not reduce SCI under the spec. Real reduction comes from the four solutions, not purchasing credits.
- Caching and batching add engineering complexity. Tradeoffs need scoping before committing.
- Model routing requires a quality validation step before going live.

Next step: Instrument the current POC for token and call counts before any optimization is applied.